

Visual Analytics Project Proposal

Francesco Mastrostefano

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1 General Idea

Multiple Sclerosis (also denoted by the acronym "MS") is one of the most common disabling neurological diseases in young adults (particularly diagnosed between 20 and 40 years old), characterized by a complex epidemiology with characters varying by multiple factors including geographic territory and the population under analysis.

The aim of this project is to develop a visual analytics application that allows users to have an intuitive and interactive analysis about epidemiology of this specific neurological disease, in particular I would like to create an application that can allow users an easy exploration of epidemiological data, analyzing it against global demographic data so that disease trends can also be extracted at the geographical region level using a simple and effective visual analysis tool.

2 Dataset (Characteristics and Context)

As dataset I want to use data downloadable from Atlas of MS, in particular from this site <https://www.atlasofms.org/map/global-of-people-with-ms> it is possible to obtain global informations of Multiple Sclerosis cases including data with respect to specific metrics, for example prevalence, incidence, demographic splits, and other indicators.

Data are collected at 3 main time points: 2008, 2013, and 2020-2022 years. To have a better, homogeneous and specific focus of Visual Analytics app I would like to focus on following metrics:

- **Country / Region**
- **Total Number of People with MS**
- **Prevalence of MS per 100,000**
- **Number of Incident Cases / Incidence per 100,000**
- **Mean Age of Onset**
- **Current Types of MS cases (RRMS, PPMS, SPMS, Unknown Disease Course)**

This choice allows me to concentrate on simpler, quantitative metrics that can be visualized effectively. Focusing on these core fields also helps reduce the complexity of the interface, making it easier to spot meaningful patterns.

In addition to MS main dataset, I'm going to join and integrate a second dataset about Smokers data over the World (downloaded from <https://ghdx.healthdata.org/record/ihme-data/gbd-2019-smoking-tobacco-use-prevalence-1990-2019>) and, analyzing it, I will obtain informations on the number of smokers per country. By combining this with MS prevalence data, I aim to explore potential correlations between smoking habit and MS cases per region. This analysis makes possible an investigation of how lifestyle factors of smokers might influence or relate to the regional distribution and severity of Multiple Sclerosis.

3 Analytics and Visual Part

3.1 Analytics Part

The main idea is to analyze MS prevalence and incidence across different countries and regions with also some other numerical factors like Population Number, Total MS cases and disease type distributions such that to have a complete overview of epidemiological data about MS disease.

Another relevant aspect of this analysis is to extract highlighting spatial and temporal patterns for a specific country or for a set of them in the data such that to analyze and explore them over the time, plus combination with smokers data to discover also a correlation between MS cases and external lifestyle factors (i.e. smoking habit).

Referring to metrics previously mentioned in dataset section, comparing them across countries and time periods, one of my goals is to reveal trends such as increasing prevalence in specific continents, outlier

countries with unusually high incidence, or demographic shifts in age for example. By combining MS cases data with smokers global data, I intend to discover whether there exist some patterns of regions with higher smoking rates also have higher MS cases.

3.2 Visual Part

The visual component of this project is designed to offer an intuitive and interactive exploration of the datasets through a set of views that are coordinated and filterable. The goal is to enable users to analyze MS data across countries, time periods, and other key indicators while also incorporating external lifestyle data such as smoking rates.

The web application includes the following main visualizations:

- **Heatmap:** This view displays the geographical distribution of MS-related metrics (such as prevalence and incidence) by shading countries according to the selected value. Users can select the year and metric to observe how the situation evolves over time.
- **Bubble Chart:** this visualization is similar to a scatter plot but with an added dimension: the bubble size. This one represents a third variable, such as population, total MS cases or a specific computed ratio. This chart is useful to highlight clusters or outliers when multiple factors are considered simultaneously.
- **Stacked Bar Chart:** using this chart I would like visualize the distribution of MS disease types (RRMS, PPMS, SPMS, Unknown) across macro-regions with also considering year of analysis. It offers a deep insight into how disease type composition varies geographically and over time.
- **Parallel Coordinates Plot:** with this chart I want to create a multi-dimensional visualization that will display selected numerical variable (e.g., number of prevalent MS cases, prevalence, mean age) for each region. Each region is represented as a line crossing all vertical axes, and this intersection represents a specific moment in time with increasing or decreasing values, making it easier to discover numerical patterns over the time.
- **PCA Clustering Plot:** in this chart are applied some clustering techniques to reduce dimensionality and group countries with similar profiles. The axes (PCA1 and PCA2) are derived from selected metrics such as total MS cases, number of smokers, mean age, and incidence or prevalence values. The result is a 2D scatter plot where colors represent clusters.

4 Intended Users

- **Public Health Researchers & Epidemiologists**
Looking to highlight and discover areas with an increasing diffusion of MS cases.
- **Healthcare Policymakers & Pharmaceutical industries**
Understanding how resources could be allocated based on data insights and how industrial production of drugs can be enhanced.
- **Clinicians & Neurological Researchers**
Aiming to discover patterns for improving clinical strategies or for starting future research projects.
- **Politicians**
Understanding and improving decision-making strategies for distributing state funds to enhance service delivery and health care.
- **Patients**
Interest on how MS is distributed globally and how this disease is evolving.

5 Various Characteristics in Relation to the Visual Analytics Cycle

- **Data Preparation**

Because the original MS dataset has many columns from Atlas of MS, I have decided to focus on a simpler subset for easier and concise visualization. As the next step, I intend to refine country/region labels correspondence and handle any missing fields to keep the data consistent, also across the various years in which the research was conducted.

- **Exploratory Analysis & Visualization**

I would like to experiment with interactive charts to see if the chosen fields reveal patterns or interesting outliers in MS cases, regions and countries, visualizing also percents of MS types.

- **Insight Generation**

Users could identify notable differences among countries or begin forming hypotheses about external influences (e.g., lifestyle, demographics).

6 Mockup (Draft)

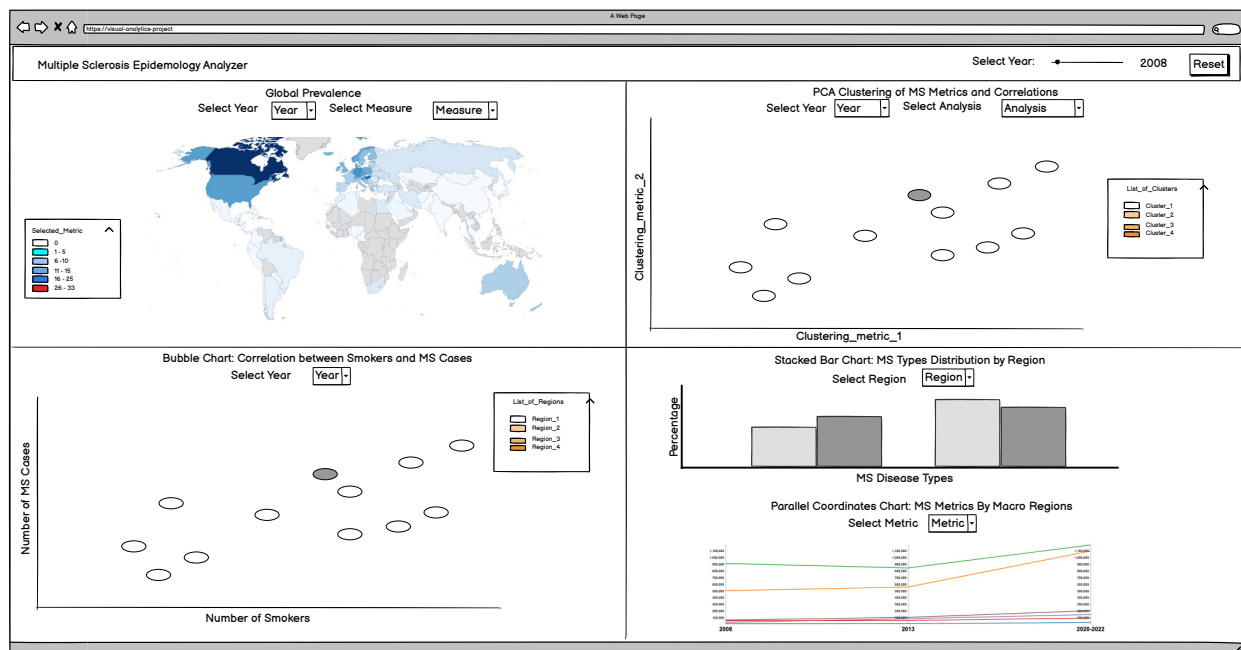


Figure 1: Mockup for the MS Visual Analytics application